CO4_AT1_DATA INTERPRETATION

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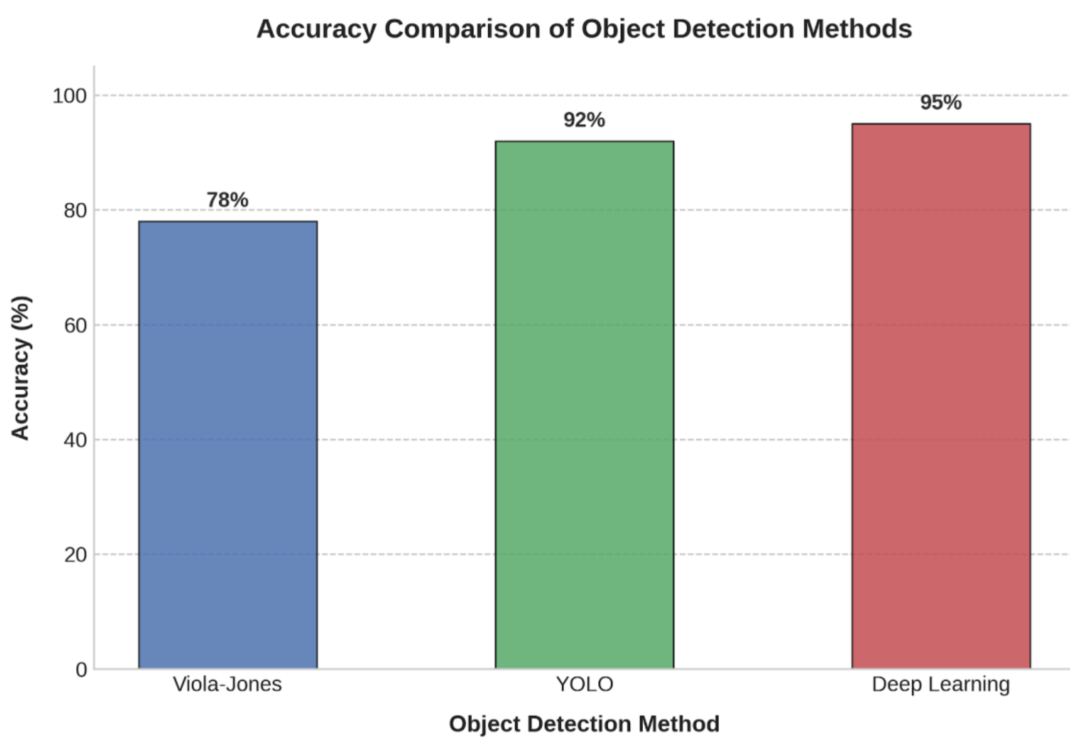
**Course : ** Computer Vision

1. Real-Time Detection System

Given Data

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

The surveillance system must operate under strict latency constraints while maintaining acceptable accuracy.



Comparison

Viola-Jones:
Viola-Jones is very fast, making it suitable for applications where latency is the most important factor. However, its accuracy is only 78%, which may not provide reliable surveillance detection.

YOLO:
YOLO provides 92% accuracy with moderate speed. It offers a good balance between detection accuracy and processing speed, making it suitable for real-time surveillance.

Deep Learning:
Deep Learning provides the highest accuracy of 95%, but its speed is slow. This can cause higher latency and may not be suitable when immediate detection is required.

Evaluation

For a surveillance system operating under strict latency constraints, YOLO should be selected. Although Viola-Jones is faster, its lower accuracy can reduce detection reliability. Deep Learning provides higher accuracy but introduces greater latency.

Conclusion

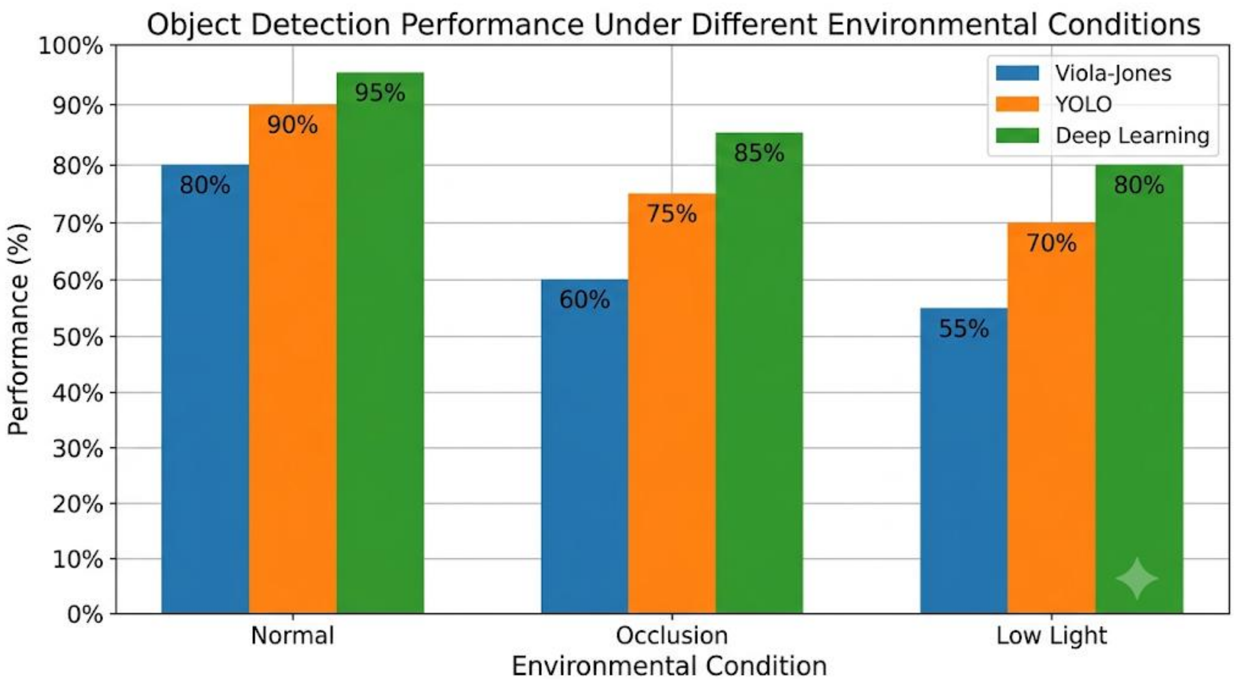
YOLO is the best choice because it provides high accuracy with moderate processing speed. It offers a practical balance between accuracy and real-time performance for surveillance deployment.

2. Robust Object Detection

Given Data

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

The values are provided in the assessment for comparing performance under normal, occluded, and low-light conditions.



Comparison

Viola-Jones:

It achieves 80% performance under normal conditions but drops to 60% under occlusion and 55% in low-light conditions. Therefore, its robustness is relatively low.

YOLO:

YOLO achieves 90% under normal conditions, 75% under occlusion, and 70% under low-light conditions. It provides better robustness than Viola-Jones.

Deep Learning:

Deep Learning achieves the highest performance in all three conditions: 95% under normal conditions, 85% under occlusion, and 80% under low-light conditions. Therefore, it provides the best overall robustness.

Evaluation

Deep Learning provides the best overall performance because it has the highest values across normal, occluded, and low-light environments. Its main trade-off is that advanced deep-learning approaches generally require greater computational resources and may increase processing requirements.

YOLO provides a useful middle ground between performance and practical deployment requirements, while Viola-Jones has lower robustness but can be useful where computational simplicity is important.

Conclusion

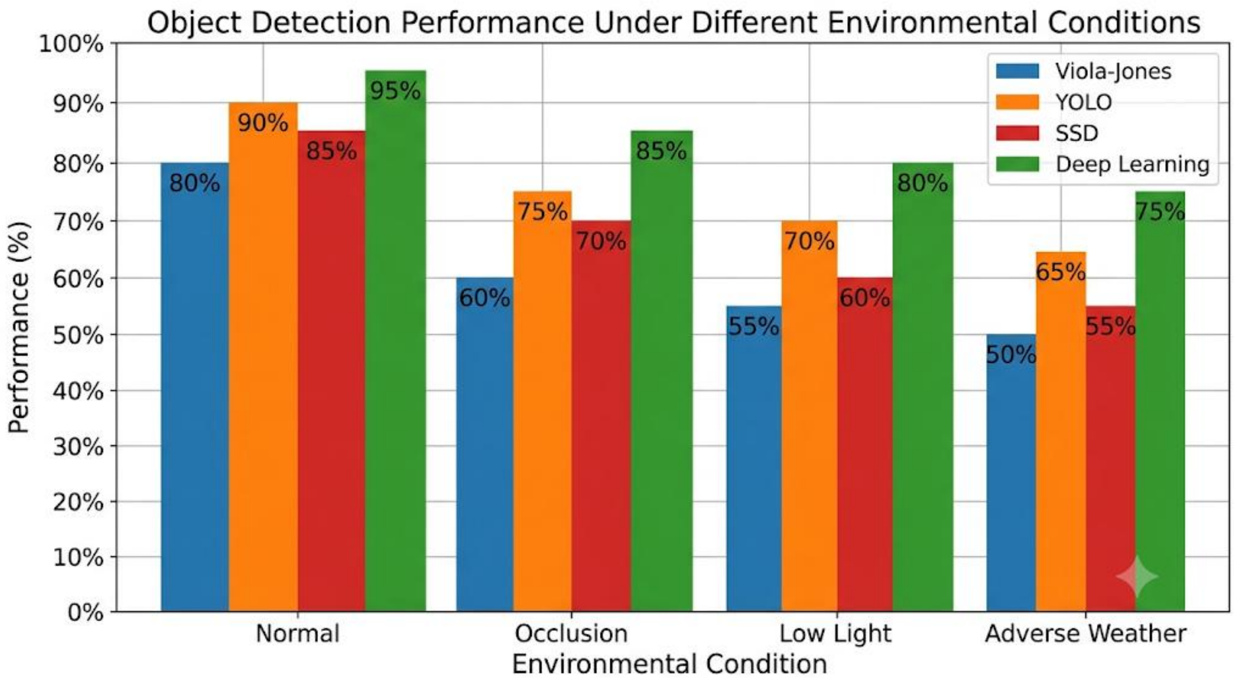
Deep Learning is the best method for overall robust object detection. It performs consistently better across all three environmental conditions. However, if computational resources and real-time processing are major constraints, YOLO may provide a more practical alternative.

3. Optical Flow Analysis

Given Information

Two optical flow methods are available:

- **Method A:** Produces stable motion vectors but misses small motions.
- **Method B:** Detects fine motion but produces noisy vectors.



Comparison

Method A:
Method A produces stable and consistent motion vectors. This makes it less sensitive to noise and suitable for applications where stable motion tracking is important. However, it may fail to detect small or subtle movements.

Method B:
Method B can detect fine and small motions, making it useful when detailed movement detection is required. However, its vectors are noisy, which can reduce tracking stability and require additional noise filtering.

Evaluation

For dynamic scenes, the choice depends on the application requirement. If the main requirement is stable and reliable tracking, Method A is preferable. If the application requires detection **of** small and fine movements, Method B is more suitable.

For general motion tracking in dynamic scenes, Method A is preferred because stable motion vectors provide more reliable tracking and are less affected by noise.

Conclusion

Method A is the better choice when accuracy and stability of overall motion tracking are more important than detecting very small movements. Method B should be selected when fine-motion detection is the primary requirement and the system can handle noisy vectors.

4. Model Generalization and Overfitting

Given Data

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

The assessment asks for comparison based on real-world reliability, generalization ability, and overfitting risk.

Model A Analysis

Model A has a training accuracy of 98%, which is very high. However, its testing accuracy falls significantly to 70%.

The difference between training and testing accuracy is:

98% − 70% = 28%

This large gap indicates that Model A has a high risk of **overfitting**. The model performs very well on training data but does not generalize well to unseen data.

Therefore, Model A is less reliable for real-world deployment.

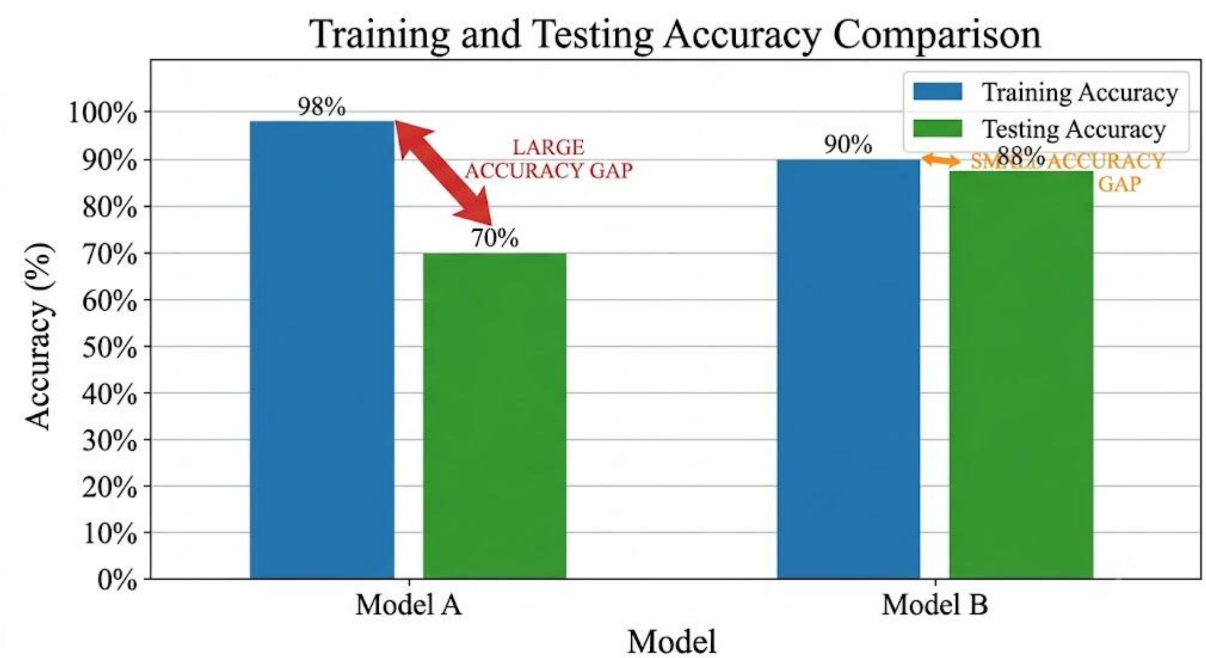
Model B Analysis

Model B has a training accuracy of 90% and testing accuracy of 88%.

The difference is:

90% − 88% = 2%

The small difference indicates that Model B performs consistently on both training and unseen testing data. Therefore, it has better **generalization ability** and a much lower risk of overfitting.



Comparison

Factor	Model A	Model B
Training Accuracy	98%	90%
Testing Accuracy	70%	88%
Accuracy Gap	28%	2%
Overfitting Risk	High	Low
Generalization	Poor	Good
Real-World Reliability	Low	High

Evaluation

Although Model A has higher training accuracy, its low testing accuracy indicates poor generalization. A model used in the real world must perform well on unseen data, not only on its training dataset.

Model B has slightly lower training accuracy but achieves much better testing accuracy and has only a 2% difference between training and testing performance.

Conclusion

Model B is more reliable for real-world deployment. It has better generalization ability, a smaller training-testing accuracy gap, and a lower risk of overfitting. Therefore, Model B should be selected despite its lower training accuracy.

5. System Performance under Constraints

Given Data

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

The task requires selecting the model that provides the best balance between accuracy, latency, and cost for a real-time application.

Model A

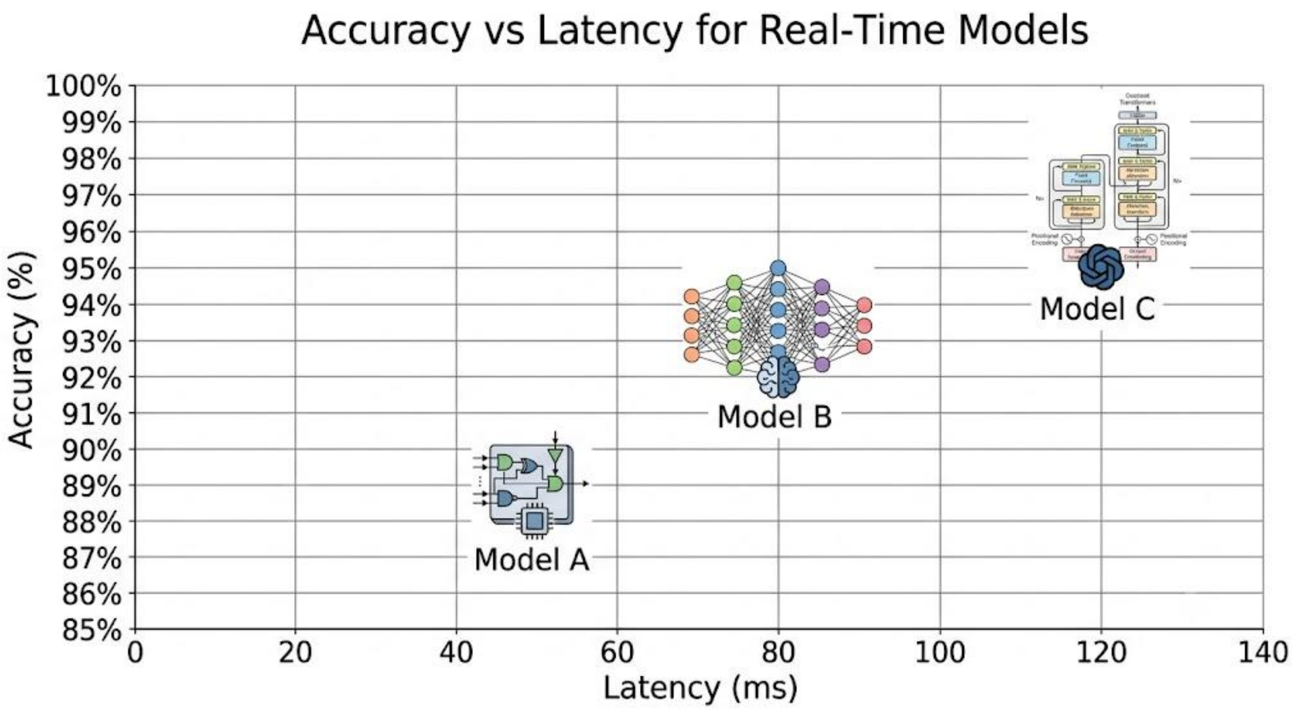
Model A has 88% accuracy, 50 ms latency, and low cost. Its low latency and cost make it attractive for real-time applications. However, its accuracy is lower than the other two models.

Model B

Model B provides 92% accuracy with 80 ms latency and moderate cost. It provides a significant improvement in accuracy compared with Model A while maintaining a moderate cost.

Model C

Model C provides the highest accuracy of 95%. However, it has the highest latency of 120 ms and high cost. The high latency may negatively affect real-time performance.



Comparison

Factor	Model A	Model B	Model C
Accuracy	88%	92%	95%
Latency	50 ms	80 ms	120 ms
Cost	Low	Moderate	High
Real-Time Suitability	High	Moderate	Low
Overall Balance	Good	**Best**	Low

Evaluation

Model A is the fastest and least expensive, but its accuracy is the lowest. Model C has the highest accuracy but also has the highest latency and cost, making it less practical for real-time deployment.

Model B provides a good compromise between accuracy, latency, and cost. It improves accuracy to 92% while keeping the cost moderate.

Conclusion

Model B provides the best overall balance for a real-time application. Although Model A has lower latency and cost, its accuracy is lower. Model C provides higher accuracy but has excessive latency and cost. Therefore, Model B is the most practical choice for real-world deployment.